

 HAMPA ENGINEERING CORPORATION HAMPA ENGINEERING & DESIGN CO (HEDCO) PROJ. NO.: 2103	 Ammonia Casale PROJ. NO.: 5158-00	LORDEGAN AMMONIA & UREA PROJECT	 شرکت پتروشیمی لردگان (سهانی عام) Lordegan Petrochemical Company
		DOC. TITLE: SECONDARY REFORMER EFFLUENT WASTE HEAT BOILER	
		C. DOC. NO.: 2103-DSH-14-PR-006 L. DOC. NO.: 5158-00-E-PSE-141	

COVER SHEET

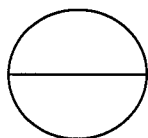
DOCUMENT TITLE: **SECONDARY REFORMER EFFLUENT WASTE HEAT BOILER**

ITEM No.: **3-E-1401**

CONTRACTOR DOC. NO.: **2103-DSH-14-PR-006**

LICENSOR DOCUMENT NO.: **5158-00-E-PSE-141**

TOTAL PAGES: **4** (INCLUDING THIS PAGE)



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REV.	DATE	DESCRIPTION	STATUS CODE	PREPARED	CHECKED	APPROVED	PROJECT



Hedco
Hampa Engineering &
Design Company

HAMPA
ENGINEERING
CORPORATION
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**LORDEGAN AMMONIA &
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EFFLUENT WASTE HEAT BOILER

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شرکت پتروشیمی لردگان (سهایی نام)
Lordegan Petrochemical Company

REV.: 1

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PROJ. NO.: 2103		PROJ. NO.: 5158-00		Item No		3-E-1401	
Service of Unit		SECONDARY REF. EFFLUENT WASTE HEAT BOILER		Connected In			
Size	mm (*) x (*)	Type	(*) HOR				
Surf/Unit (Eff)	(*) m2	Shell/Unit	ONE	Surf/Shell (Gross/Eff) (*) m2			
Fluid Allocation		Shell Side		Tube Side			
Fluid Name		BOILER FEED WATER AND STEAM		SECONDARY REFORMER EFFLUENT			
Fluid Quantity, Total		(*)		279'566 x 1.1			
Vapor (In/Out)				176'878 x 1.1		176'878 x 1.1	
Liquid							
Steam		316'130 x 1.1		102'688 x 1.1		102'688 x 1.1	
Water		(*)		(*)			
Noncondensable							
Temperature (In/Out)		323.3		323.3		967.5 355.0	
Density v / l		kg/m3		5.36		10.48	
Molecular Weight, Vapor		kg/kmol		15.99		15.99	
Mol. Weight, Noncondensable							
Specific Heat Vapour		kJ/kg-C		2.353		2.119	
Specific Heat Liquid							
Thermal Conduct Vapour		W/m-C		0.1722		0.1011	
Thermal Conduct Liquid							
Viscosity Vapour		cP		0.0376		0.0223	
Viscosity Liquid							
Latent Heat		kJ/kg					
Inlet Pressure		bar a		117.84 (12)		34.74	
Velocity		m/sec					
Pressure Drop, Allow/Calc		bar		0.3 (7)		(*)	
Fouling Resistance (min)		m2 °C/W		0.00015		0.00050 	
Heat Exchanged		MW		106.013 x 1.1 (7)		MTD (Corrected) (*) C	
Transfer Rate, Service		W/m2 °C		(*)		Clean (*)	
CONSTRUCTION OF ONE SHELL				Sketch (Bundle/Nozzle Orientation)			
Design/Test Pressure		bar g		130 (8) / 38 /			
Design Temperature		C		335		(4)(5)	
No Passes per Shell				one		one	
Corrosion Allowance		mm		3		3	
Connections In		STUB END (*)		1500 mm ID WELD STUB + INTERNAL REFRACTORY LINING (1000 mm ID)			
Size & Out		STUB END (*)		28" OD STUB END			
Rating Intermediate							
Tube No	(*)	OD mm	(*)	Thk(Avg)	(*)	mm	L. (*) mm Pitch (*) mm Layout (*)
Tube Type		(*)		Material SA-213-T11 (2)			
Tubes Seamless							
Shell ID	(*)	mm	SA-516-70	Thk	mm	Shell Cover	
Channel or Bonnet		SA-387-11-CL2 (5)				Channel Cover SA-387-11-CL2 (5)	
Tubesheet-Stationary		SA-182-F11 INLET W/REFRAC LINING				Tubesheet-Floating	
Floating Head Cover						Impingement Plate	
Baffles-Cross	C.S.	Type	(*)	%Cut (Diam)	Spacing(c/c) mm		Inlet mm
Baffles-Long				Seal Type			
Supports-Tube				U-Bend		Type	
Bypass Seal Arrangement				Tube-Tubesheet Joint		STRENGTH WELD	
Expansion Joint				Type		Insulation YES, TYPE "H" (9)	
Rho-V2-Inlet Nozzle (kg/m-sec2)				Bundle Entrance		Bundle Exit kg/m-sec2	
Gaskets-Shell Side				Tube Side			
-Floating Head							
Code Requirements		ASME VIII, DIV-1		ASME STAMP REQUIRED		TEMA Class R where applicable	
Radiography XR		Shell		Heads			
Stress relieving		Shell		Heads			
Weight/Shell		Filled with Water		Bundle		kg	
Notes (1) (*) SELLER SHALL STATE IN QUOTATION. (2) FERRULES INCONEL 601 AT INLET SIDE ONLY. (3) HYDROGEN SERVICE ON TUBESIDE, H2 PARTIAL PRESSURE IS 12.80 BAR A. (4) METAL DESIGN TEMPERATURE INLET & OUTLET CHANNEL = 410°C.							
(5) REFRACTORY LINE INLET CHANNEL FOR A MAXIMUM METAL TEMPERATURE OF 200 °C CONSIDERING OUTSIDE STILL AIR AT 44 °C. DESIGN INTERNAL TEMPERATURE AT GAS INLET SIDE IS 1010 °C. 							
(6) PROVIDE INTERNAL BY-PASS W/PNEUMATICALLY OPERATED VALVE.							
(7) 10% OVERDESIGN IS REQUIRED ON FLOWS AND DUTY.							
(8) DESIGN FOR FV, DESIGN PRESSURE IS 130 BAR G PLUS MAX. STATIC HEAD (HILL INTO THE STEAM DRUM) BEING STEAM DRUM DESIGNED AT 130 BAR G.							
(9) INSULATION SUPPORTS REQUIRED ON SHELLSIDE AND ON OUTLET CHANNEL. (10) PAINTING REQUIRED.							
(11) MINIMUM DESIGN METAL TEMPERATURE IS -27 °C.							
(12) PRESSURE AT STEAM DRUM: STATIC HEAD TO BE ADDED.							
(13) HYDRAULIC CALCULATION SHEET OF BOILER WATER CIRCULATION TO BE SUBMITTED BY VENDOR.							

